Mahmoud Alyosify

AI Security Engineer | Threat Intelligence, OSINT Automation & ML for Security
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PROFESSIONAL SUMMARY

AI Security Engineer applying machine learning to **threat intelligence, OSINT automation and media authentication**. Trained in **AI for Cybersecurity at NTI (420 hours)** and applied it inside **e& Egypt** across threat intelligence, digital forensics and enterprise security analysis. Built **HORUS Sentinel**, an autonomous multi-agent OSINT platform that correlates continuously collected open-source intelligence into a **living knowledge graph** and generates **prioritised, evidence-backed reports** from a self-hosted fine-tuned LLM — compressing hours of analyst work into minutes. Also fine-tuned **Meta-Llama-3-8B** on **159,826 GTD/GDELT records** for global event intelligence, and built a **CNN deepfake-audio detector** for fraud detection and digital forensics. Currently an **AI Engineer Intern at Injaz Tech**, hardening an LLM assistant with prompt-injection guardrails, rate limiting and hallucination checks. Completing an **MSc in Artificial Intelligence at Queen's University, Canada** (expected **Nov 2026**) on a **Digilians Presidential Scholarship**.

EDUCATION

MSc in Artificial Intelligence — *Sep 2025 – Nov 2026*
Queen's University, Kingston, Ontario, Canada — Expected Nov 2026 · GPA: 4.30 / 4.30

- **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
- **Master's Thesis:** *MinimalLM: Verbosity-Aware Decoding for Token-Efficient Language Model Inference* — Supervisor: Dr. Rahatara Ferdousi.
- Coursework: **Large Language Models, Generative AI, Deep Learning, Cloud Computing, End-to-End ML Pipelines**.

B.Sc. Computer & Information Science (Bioinformatics) — *Sep 2019 – Jul 2023*
Assiut University, Egypt — GPA: 3.53 / 4.00

- Graduation Project: **Vitalism Solution** — **Grade A+**; **Semi-Finalist, Microsoft Imagine Cup 2023**.

TECHNICAL SKILLS

Security & Intelligence	Open-Source Intelligence (OSINT), Cyber Threat Intelligence, Digital Forensics, Enterprise Security Analysis, Event & Threat Data (GTD, GDELT), Intelligence Knowledge Graphs, Automated Reporting, Evidence Traceability, Prompt-Injection Defence & LLM Guardrails, Rate Limiting
ML for Security	Deepfake & Synthetic-Media Detection, Anomaly & Rare-Event Detection, Imbalanced Classification, Risk Prediction, Adversarial ML Concepts, CNNs, Spectrogram-Based Classification
AI Systems	Multi-Agent Systems, Autonomous Agents, Fine-Tuning (LoRA, QLoRA, PEFT), RAG, Self-Hosted Inference, LLM APIs, Prompt Engineering, LangChain, LangGraph, FAISS, ChromaDB
Engineering & Cloud	Python, PyTorch, TensorFlow, Scikit-learn, Pandas, AWS, Docker, FastAPI, Streamlit, ONNX, SQL, MongoDB, Git/GitHub, Linux
Programming	Python, SQL, C++, C, C#, Go, JavaScript, Bash

PROFESSIONAL EXPERIENCE

AI Engineer Intern — *Oct 2026 – Present*
Injaz Tech — Injaz Group (formerly Fingerprint Consultancy) — Maadi, Cairo, Egypt

- Hardened an LLM assistant with a **prompt-injection filter**, output leak detection, per-session and per-IP **rate limiting** and refusal over hallucination.
- Red-teamed it with injection probes: **0** system-prompt leaks in 8 runs.

Part-Time Instructor — Machine Learning & AI for Business — *Jul 2025 – Present*
National Telecommunication Institute (NTI) — Egypt

- Teach **Machine Learning** (90-hour bootcamps) and **AI for Business** modules in NTI's Summer Internships Program, on separate engagements.

AI in Cybersecurity Intern — *Jun 2025 – Jul 2025*
e& Egypt — via NTI (On-the-Job Training) — Cairo, Egypt

- Applied AI-driven security tooling across **threat intelligence, digital forensics** and **enterprise security analysis** inside a national telecommunications operator.
- Completed the accompanying **420-hour AI in Cybersecurity** program at NTI covering applied AI methods for security operations.

Systems Engineer (Military Service) *Feb 2024 – Mar 2025*
National Company for Roads Building & Development Cairo, Egypt

- Supported backend systems for **national critical infrastructure** — toll systems serving **50%+ of Egyptian vehicles** across **70%+ of national roads** — in the Control and Command environment.
- Co-developed an internal data-management and reporting application in **C#, .NET, Entity Framework, SQL Server**; supported legacy systems to maintain operational continuity.

Online Instructor — AI, Algorithms & Data Science *2020 – Present*
Udemy & YouTube Remote

- Built Arabic-language technical curricula reaching **10,000+ learners** and **700K+ YouTube views**.

SELECTED SECURITY & INTELLIGENCE PROJECTS

HORUS Sentinel — Autonomous Multi-Agent OSINT & Threat-Intelligence Platform *2026*
Python · Multi-Agent Systems · Fine-Tuned LLM · Knowledge Graphs · Self-Hosted Inference

- Architected an autonomous intelligence analyst: a **swarm of specialised passive collection agents** continuously gathers open-source intelligence without active probing of targets.
- Correlated every finding into a **living Intelligence Knowledge Graph**, so signals are linked across sources and over time rather than assessed in isolation.
- A **self-hosted, fine-tuned language model** reasons over the graph to produce **prioritised, evidence-backed intelligence reports** with traceable sourcing — keeping sensitive data inside the environment rather than sending it to a third-party API.

Horus-OSINT — Cloud-Based Threat-Intelligence Assistant *2026*
Meta-Llama-3-8B · Fine-Tuning · GTD / GDELT · AWS

- Fine-tuned **Meta-Llama-3-8B** on **159,826** curated **GTD** and **GDELT** records covering global terrorism and event data.
- Deployed on **AWS** as an interactive assistant that lets analysts query global event intelligence in natural language instead of writing structured queries.

AudioShield — Deepfake Audio Detection for Forensics & Fraud Prevention *2023*
TensorFlow · CNNs · Librosa · Streamlit

- Trained a **CNN** on **mel-spectrograms** to distinguish authentic from synthetic speech, deployed with real-time prediction and automated PDF reporting.
- Targeted at **voice-fraud detection, digital forensics** and **media authentication** — an increasingly common social-engineering vector.

Analyzing Review Effort: Human vs. Agentic Pull Requests *2026*
Repository Mining · Code-Quality Risk · Empirical Analysis

- Quantified review burden and recurring failure modes in AI-generated versus human-authored code — directly relevant to supply-chain and code-integrity risk as coding agents enter development pipelines.

Additional Projects

- **SCRAP** — rare-event risk prediction on imbalanced data, forecasting satellite collision risk **48+ hours** ahead.
- **MinimalLM** — MSc thesis: Fewer generated tokens are a real saving only when the answer is still complete. MinimalLM gives both, without training. A training-free decoding method with completeness guardrails makes three open LLMs generate **47.6%** fewer tokens, with no answer cut off.
- **Also:** SimCLR-Vision-SSL · RL-Cloud-Autoscaler · Automated Multimodal Agent · Vitalism Solution.

CERTIFICATIONS & AWARDS

AI in Cybersecurity — NTI (420 hrs)
AWS Academy — ML Foundations, ML for NLP, Data Engineering
ML Internship — ITIDA / Egypt Makes Electronics (200 hrs)
1st Place — Smart Cities Hackathon 2022
HCIP – AI (Huawei)
Machine Learning Specialization — DeepLearning.AI
Microsoft Certified: Data Analyst Associate
Semi-Finalist — Microsoft Imagine Cup 2023

LANGUAGES

- **Arabic:** Native · **English:** Professional Working Proficiency.